2025 SSMO Relay Round 1 Solutions

SMO Team

RR 1 Part 1: Let $x_1, x_2, \ldots, x_7$ be distinct integers such that the mean of $\{x_i, x_{i+1}, x_{i+2}\}$ is an integer for all integers $1 \leq i \leq 5$. Find the minimum possible positive value of $x_7 - x_1$.

Answer: $\boxed{3}$

Solution: Since the mean of each three consecutive terms is an integer, $x_i + x_{i+1} + x_{i+2}$ is divisible by 3 for all $1 \leq i \leq 5$. So,

$$(x_{i+1} + x_{i+2} + x_{i+3}) - (x_i + x_{i+1} + x_{i+2}) = x_{i+3} - x_i$$

is divisible by 3 for all $1 \leq i \leq 4$, meaning $x_7 \equiv x_4 \equiv x_1 \pmod 3$. Thus, $x_7 - x_1$ is a multiple of 3. Since $(x_1, x_2, \ldots, x_7) = (0, 1, 2, 6, 4, 5, 3)$ satisfies the conditions, the answer is $\boxed{3}$.

RR 1 Part 2: Let $T = TNYWR$. A positive integer is called *zro* if more than half of its digits are 0. Find the sum of the first T^2 zro numbers.

Answer: $\boxed{4500}$

Solution: We have $T^2 = 9$. Since the leading digit of a positive integer is nonzero, no one-digit or two-digit number is zro. A three-digit number is zro if and only if its last two digits are both 0, so the first 9 zro numbers are $100, 200, \ldots, 900$. So, our answer is

$$100 + 200 + \cdots + 900 = 100 \cdot \frac{9 \cdot 10}{2} = \boxed{4500}.$$

RR 1 Part 3: Let $T = TNYWR$. Positive integers m and n satisfy $m^2 - n^2 = T$. What is the least possible value of $m + n$?

Answer: $\boxed{90}$

Solution: Note that $(m + n)(m - n) = T = 4500$. Since $m + n$ and $m - n$ have the same parity and 4500 is even, they must both be even. To minimize $m + n$, we want the two factors to be as close as possible, and the closest such pair is $90 \cdot 50$. This gives $(m, n) = (70, 20)$, so the answer is $\boxed{90}$.

2025 SSMO Relay Round 2 Solutions

SMO Team

RR 2 Part 1: Let (p, q, r) be the real solution to the following system:

$$\begin{aligned} pq &= 20 \\ qr &= 25 \\ r + p &= 9. \end{aligned}$$

Compute $p^2 + q^2 + r^2$.

Answer: $\boxed{66}$

Solution: Dividing the first equation by the second, we have $\frac{p}{r} = \frac{4}{5}$. Combined with $r + p = 9$, we have $p = 4$ and $r = 5$, so $q = 5$. Thus,

$$p^2 + q^2 + r^2 = 16 + 25 + 25 = \boxed{66}.$$

RR 2 Part 2: Let $T = TNYWR$. Let $x_1, x_2, \ldots, x_T$ be an increasing sequence of positive integers such that for every positive integer $1 \leq n \leq T$, the sum $x_1 + x_2 + \cdots + x_n$ is a multiple of n. Find the smallest possible value of $x_T - x_1$.

Answer: $\boxed{130}$

Solution: Let $a_n = \frac{x_1 + x_2 + \cdots + x_n}{n}$, which is an integer for all $1 \leq n \leq T$. Since the sequence is increasing, x_{n+1} is greater than each of $x_1, x_2, \ldots, x_n$, so $x_{n+1} > a_n$. This means

$$a_{n+1} = \frac{na_n + x_{n+1}}{n+1} > a_n \implies a_{n+1} \geq a_n + 1,$$

so $a_T \geq a_1 + (T-1) = x_1 + (T-1)$. Now,

$$x_T = Ta_T - (T-1)a_{T-1} = a_T + (T-1)(a_T - a_{T-1}) \geq x_1 + 2(T-1).$$

Equality holds when $x_n = 2n - 1$, since $1 + 3 + \cdots + (2n-1) = n^2$. So, our answer is $2(T-1) = 2 \cdot 65 = \boxed{130}$.

RR 2 Part 3: Let $T = TNYWR$. Define a *multiplicative partition* of a positive integer s as the value of a product $a_1 a_2 \cdots a_k$, where $a_1 + a_2 + \cdots + a_k = s$ and every a_i is a positive integer. Let $f(x)$ denote the maximal possible value of a multiplicative partition of x. If the sum of all possible values of $\frac{f(x+\lfloor\sqrt{T}\rfloor)}{f(x)}$ for integers $x > 1$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers, find $m + n$.

Answer: $\boxed{463}$

Solution: We have $\left\lfloor \sqrt{T} \right\rfloor = \lfloor \sqrt{130} \rfloor = 11$. For $x > 1$, a maximal multiplicative partition never uses a 1, and any $a_i \geq 5$ can be replaced by 2 and $a_i - 2$, since $2(a_i - 2) > a_i$. In addition, a 4 can be replaced by $2 \cdot 2$, and three 2's can be replaced by two 3's, since $2^3 < 3^2$. So, $f(x)$ uses as many 3's as possible, giving

$$f(x) = \begin{cases} 3^{x/3} & x \equiv 0 \pmod 3 \\ 4 \cdot 3^{(x-4)/3} & x \equiv 1 \pmod 3 \\ 2 \cdot 3^{(x-2)/3} & x \equiv 2 \pmod 3. \end{cases}$$

Since $11 \equiv 2 \pmod 3$, we have

$$\begin{aligned} \frac{f(x+11)}{f(x)} &= \frac{2 \cdot 3^{(x+9)/3}}{3^{x/3}} = 54 && \text{for } x \equiv 0 \pmod 3, \\ \frac{f(x+11)}{f(x)} &= \frac{3^{(x+11)/3}}{4 \cdot 3^{(x-4)/3}} = \frac{243}{4} && \text{for } x \equiv 1 \pmod 3, \\ \frac{f(x+11)}{f(x)} &= \frac{4 \cdot 3^{(x+7)/3}}{2 \cdot 3^{(x-2)/3}} = 54 && \text{for } x \equiv 2 \pmod 3. \end{aligned}$$

So, the sum of all possible values is

$$54 + \frac{243}{4} = \frac{459}{4} \implies 459 + 4 = \boxed{463}.$$

2025 SSMO Relay Round 3 Solutions

SMO Team

RR 3 Part 1: Michael, Nico, and Jackson are splitting a 100 dollar bill. Michael's favorite number is 3, so he insists on paying a positive amount of dollars that is a multiple of 3. Nico and Jackson are both scared of even numbers, so they each insist on paying an odd number of dollars. In how many ways can they pay the bill?

Answer: $\boxed{392}$

Solution: Let Michael, Nico, and Jackson pay m, n, and j dollars, respectively. Since n and j are both odd, $m = 100 - n - j$ is even, so m is a multiple of 6. For $m = 6k$, where $1 \leq k \leq 16$, there are $\frac{100-6k}{2} = 50 - 3k$ ways to write $100 - 6k$ as the sum of two positive odd integers n and j. So, the answer is

$$\sum_{k=1}^{16}(50 - 3k) = 16 \cdot 50 - 3 \cdot \frac{16 \cdot 17}{2} = 800 - 408 = \boxed{392}.$$

RR 3 Part 2: Let $T = TNYWR$. In triangle ABC, angle C is a right angle. If the median from C has length T, find the maximum possible area of triangle ABC.

Answer: $\boxed{153664}$

Solution: Since $\angle ACB = 90°$, C lies on the circle with diameter AB, and the median from C is a radius of this circle. So, $AB = 2T = 784$, and the distance from C to AB is at most the radius, 392. Thus, the maximum possible area is

$$\frac{784 \cdot 392}{2} = \boxed{153664},$$

which occurs when $CA = CB$.

RR 3 Part 3: Let $T = TNYWR$. The numbers 1 and 2 are written on a board. Every second, if the numbers currently on the board are x and y, the number $x+y$ is written, and one of x or y is erased. After a finite number of seconds, the larger number on the board is $\left\lfloor \sqrt{T} \right\rfloor$. How many possible values can the smaller number take?

Answer: $\boxed{168}$

Solution: We have $\left\lfloor \sqrt{T} \right\rfloor = \lfloor \sqrt{153664} \rfloor = 392$. Note that $\gcd(x + y, x) = \gcd(x + y, y) = \gcd(x, y)$, so the two numbers on the board are always relatively prime. So, the smaller number must be relatively prime to 392. Conversely, suppose that $a > b$ are relatively prime. If $b = 1$, then the board reaches 1 and a by repeatedly erasing the larger number, starting from 1 and 2. Otherwise, $a - b$ and b are distinct and relatively prime, and the board reaches a and b from $a - b$ and b by writing a and erasing $a - b$. So, by induction on

$a+b$, every such pair can be reached. Thus, the smaller number can be any positive integer less than 392 that is relatively prime to $392 = 2^3 \cdot 7^2$, giving

$$\varphi(392) = 392 \cdot \frac{1}{2} \cdot \frac{6}{7} = \boxed{168}$$

possible values.

2025 SSMO Relay Round 4 Solutions

SMO Team

RR 4 Part 1: Call a positive integer *chuzzed* if the sum of the digits in its binary representation is equal to the units digit of its base-10 representation. Similarly, call a positive integer *chopped* if its binary representation does not contain two consecutive ones. Find the number of positive integers less than 128 that are chuzzed and chopped.

Answer: $\boxed{3}$

Solution: Note that every positive integer less than 128 has at most 7 binary digits, so a chopped number has at most 4 ones. We will do casework on the number of ones k. If $k = 1$, the number is a power of 2, and the only one with units digit 1 is 1. If $k = 2$, the chopped numbers are

$$5, 9, 10, 17, 18, 20, 33, 34, 36, 40, 65, 66, 68, 72, 80,$$

of which only 72 has units digit 2. If $k = 3$, the chopped numbers are

$$21, 37, 41, 42, 69, 73, 74, 81, 82, 84,$$

of which only 73 has units digit 3. Finally, if $k = 4$, the only chopped number is $1010101_2 = 85$, which does not have units digit 4. So, the answer is $\boxed{3}$.

RR 4 Part 2: Let $T = TNYWR$. Jonathan and Kate are playing a game with n sticks. On each turn, a player may remove 1, 2, or 3 sticks. The player who picks up the last stick loses. Kate is first to remove sticks, and both players play optimally. For how many values of n in the range $\left[T^3, 2T^3\right]$ does Kate have a winning strategy?

Answer: $\boxed{21}$

Solution: We have $T^3 = 27$. Note that Kate loses exactly when $n \equiv 1 \pmod 4$. If $n \equiv 1 \pmod 4$, then whenever Kate removes k sticks, Jonathan can remove $4 - k$ sticks, keeping the number of sticks 1 more than a multiple of 4, until Kate is forced to pick up the last stick. Otherwise, Kate can remove 1, 2, or 3 sticks to leave 1 more than a multiple of 4, and use this strategy herself. There are $54 - 27 + 1 = 28$ integers in $[27, 54]$, and 7 of them $(29, 33, \ldots, 53)$ are 1 more than a multiple of 4. So, the answer is $28 - 7 = \boxed{21}$.

RR 4 Part 3: Let $T = TNYWR$. A particle moves in the coordinate plane such that at any time t, its position is

$$\left(\sum_{a=1}^{T-1} \cos(at), \sum_{a=1}^{T-1} \sin(at)\right).$$

Over the time interval $t \in (0, k]$, the particle lies on a coordinate axis at T distinct times. The minimal value of k can be expressed as $\frac{m\pi}{n}$, where m and n are relatively prime positive

integers. Find $m+n$.

Answer: $\boxed{17}$

Solution: Let $z = e^{it}$. The position of the particle corresponds to the complex number

$$\sum_{a=1}^{T-1} z^a = \frac{z\left(z^{T-1}-1\right)}{z-1} = \frac{z^{\frac{T+1}{2}}\left(z^{\frac{T-1}{2}} - z^{-\frac{T-1}{2}}\right)}{z^{\frac{1}{2}}\left(z^{\frac{1}{2}} - z^{-\frac{1}{2}}\right)} = e^{\frac{iTt}{2}} \cdot \frac{\sin\left(\frac{(T-1)t}{2}\right)}{\sin\left(\frac{t}{2}\right)}.$$

So, the particle lies on a coordinate axis if and only if $\sin\left(\frac{(T-1)t}{2}\right) = 0$ (the particle is at the origin) or $\frac{Tt}{2}$ is a multiple of $\frac{\pi}{2}$. This means t is a multiple of $\frac{2\pi}{T-1} = \frac{\pi}{10}$ or $\frac{\pi}{T} = \frac{\pi}{21}$, and these only coincide at multiples of π. For $0 < t \leq \frac{7\pi}{10}$, there are $\left\lfloor \frac{7 \cdot 21}{10} \right\rfloor = 14$ multiples of $\frac{\pi}{21}$ and 7 multiples of $\frac{\pi}{10}$, for a total of 21 times. Since the last of these is $t = \frac{7\pi}{10}$, the minimal value of k is

$$\frac{7\pi}{10} \implies 7 + 10 = \boxed{17}.$$

2025 SSMO Relay Round 5 Solutions

SMO Team

RR 5 Part 1: The numbers 17 and 25 are written on a blackboard. Every second, if the number n is currently written on the blackboard, it is replaced by $3n$ with probability p and by $2n+3$ with probability $1-p$. For example, after one second, the two numbers on the blackboard may be $3 \cdot 17 = 51$ and $2 \cdot 25 + 3 = 53$. Given that the expected value of the number descended from 25 minus the number descended from 17 after six seconds is 1000, the value of p can be expressed as $\sqrt{a}-b$, where a and b are positive integers. Find $a+b$.

Answer: $\boxed{7}$

Solution: Note that if n is currently written on the blackboard, then the expected value of the number that replaces it is

$$p \cdot 3n + (1-p)(2n+3) = (2+p)n + 3(1-p).$$

So, from linearity of expectation, the expected difference between the two numbers is multiplied by $2+p$ every second. After six seconds, the expected difference is $(25-17)(2+p)^6 = 8(2+p)^6$, giving

$$8(2+p)^6 = 1000 \implies (2+p)^2 = 5 \implies p = \sqrt{5}-2 \implies 5+2 = \boxed{7}.$$

RR 5 Part 2: Let $T = TNYWR$. Ethan places T red cards and T blue cards in a row so that the leftmost card is red and every second card is red. He performs a sequence of operations called *replacements*: in each replacement, he selects a card other than the leftmost one uniformly at random, discards it, and moves the leftmost card to the discarded card's position. This process is repeated until only one card remains. The probability that the final remaining card is red can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

Answer: $\boxed{20}$

Solution: We claim that for a row of $n \geq 2$ cards $c_1, c_2, \ldots, c_n$, the final card is equally likely to be any of $c_1, c_2, \ldots, c_{n-1}$, and is never c_n. We proceed by induction on n. The base case $n = 2$ holds, since c_2 is discarded. Now, suppose the first replacement discards c_j. If $j = n$, then the new row is $c_2, \ldots, c_{n-1}, c_1$, and if $j < n$, then c_1 takes the place of c_j and c_n is still the rightmost card. In both cases, c_n cannot be the final card. From the inductive hypothesis, c_1 is the final card with probability

$$\frac{n-2}{n-1} \cdot \frac{1}{n-2} = \frac{1}{n-1},$$

since it is only eligible when $j < n$. For $2 \leq k \leq n-1$, c_k is eligible whenever $j \neq k$, so it is the final card with probability

$$\frac{n-2}{n-1} \cdot \frac{1}{n-2} = \frac{1}{n-1}.$$

This completes the induction. Now, there are $2T$ cards, and since every second card is red starting from the leftmost card, the rightmost card is blue. Of the other $2T-1$ cards, T are red, so the probability is

$$\frac{T}{2T-1} = \frac{7}{13} \implies 7+13 = \boxed{20}.$$

RR 5 Part 3: Let $T = TNYWR$. How many positive integers n satisfy

$$\text{lcm}(\gcd(n, n^2+T^2), \gcd(T, n^2+T^2)) = \text{lcm}(n, T)?$$

Answer: $\boxed{8}$

Solution: Note that

$$\gcd(n, n^2+T^2) = \gcd(n, T^2) \quad \text{and} \quad \gcd(T, n^2+T^2) = \gcd(T, n^2).$$

Since both of these divide T^2, so does the left hand side, meaning $n \mid \text{lcm}(n, T) \mid T^2$. So, $\gcd(n, T^2) = n$, and the equation becomes

$$\text{lcm}\left(n, \gcd(T, n^2)\right) = \text{lcm}(n, T).$$

For each prime $p \mid T$, let $t = v_p(T)$ and $e = v_p(n)$, where $0 \le e \le 2t$. Comparing the powers of p, we need

$$\max(e, \min(t, 2e)) = \max(e, t),$$

which holds if and only if $2e \ge t$. So, $v_p(n)$ can be any integer from $\lceil \frac{t}{2} \rceil$ to $2t$. Since $T = 20 = 2^2 \cdot 5$, there are 4 choices for $v_2(n)$ and 2 choices for $v_5(n)$, giving $4 \cdot 2 = \boxed{8}$ possible values of n.